REGENERATIVE BRAKING: STORING THE BACK EMF OF A BLDC MOTOR USING PWM SIGNAL

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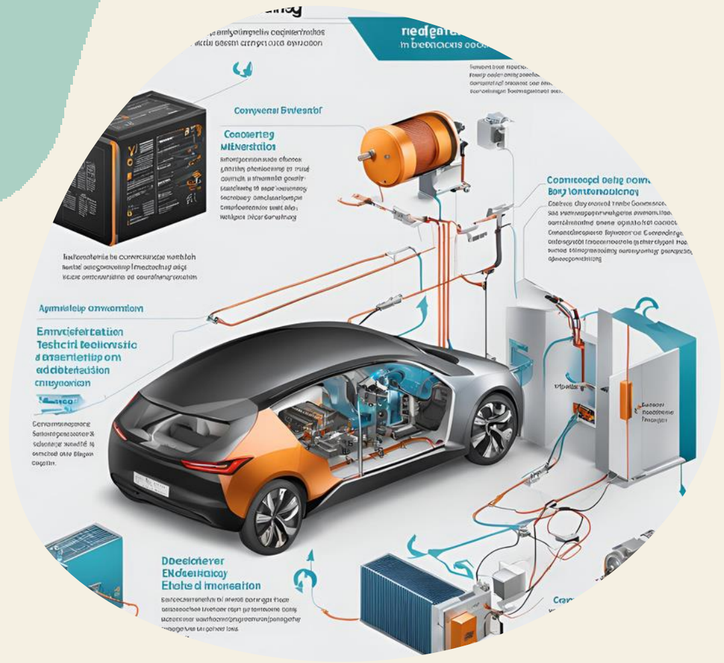


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Introduction



- Electric Vehicles have taken the automobile market by storm due to their versatility.
- Electric Vehicles can store the lost kinetic energy during braking into a battery pack - Regenerative Breaking
- We aim to create a prototype that demonstrates this mechanism

Basic Idea

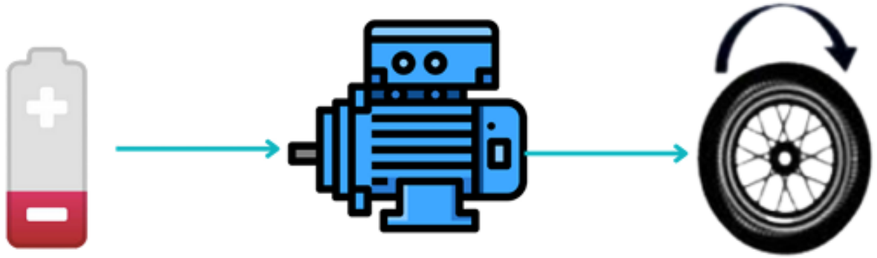


Fig: Driving Mode

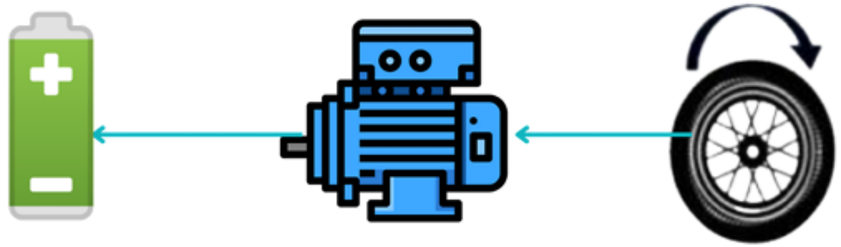
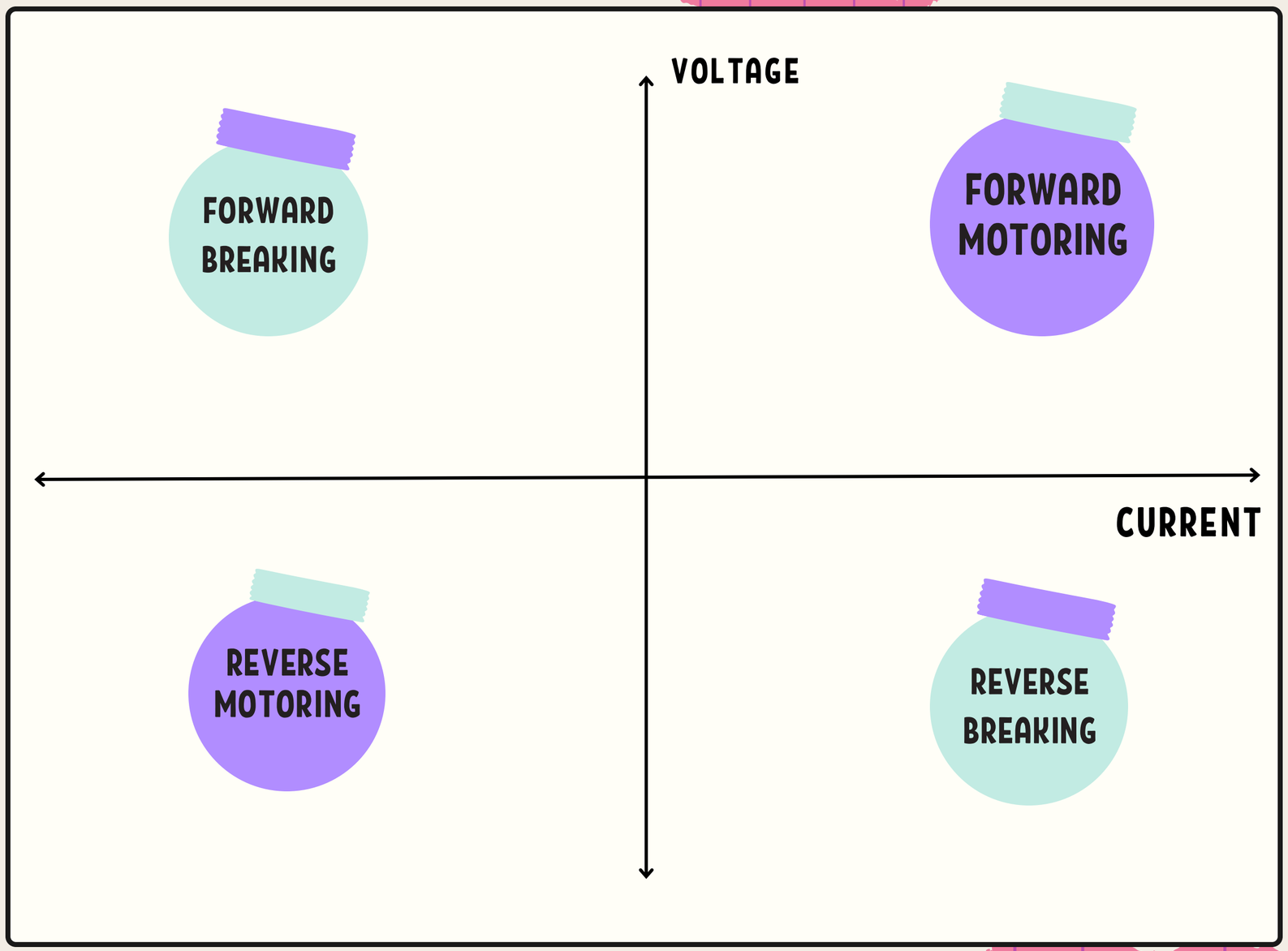


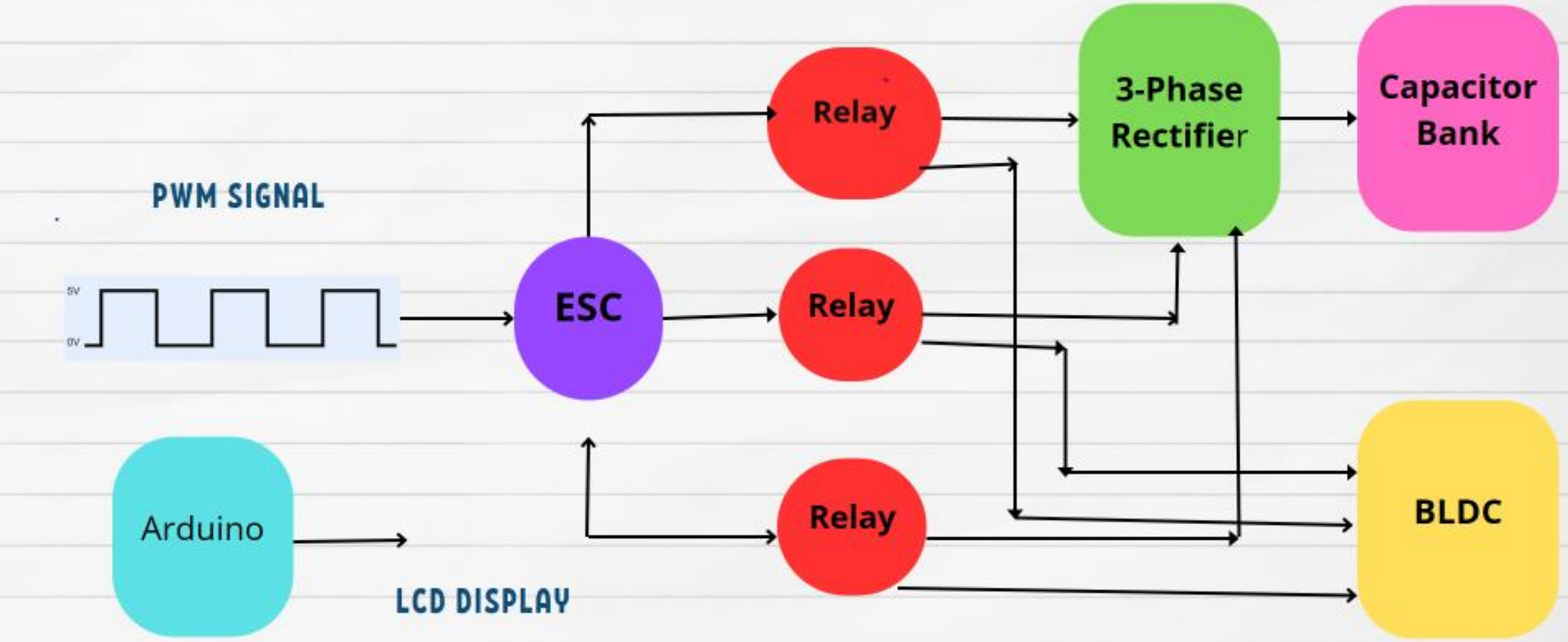
Fig: Regenerative Mode

MOTOR OPERATING MODES

1. FORWARD MOTORING
2. REVERSE MOTORING
3. FORWARD BREAKING
4. REVERSE BREAKING



BLOCK DIAGRAM OF THE SYSTEM



COMPONENTS USED













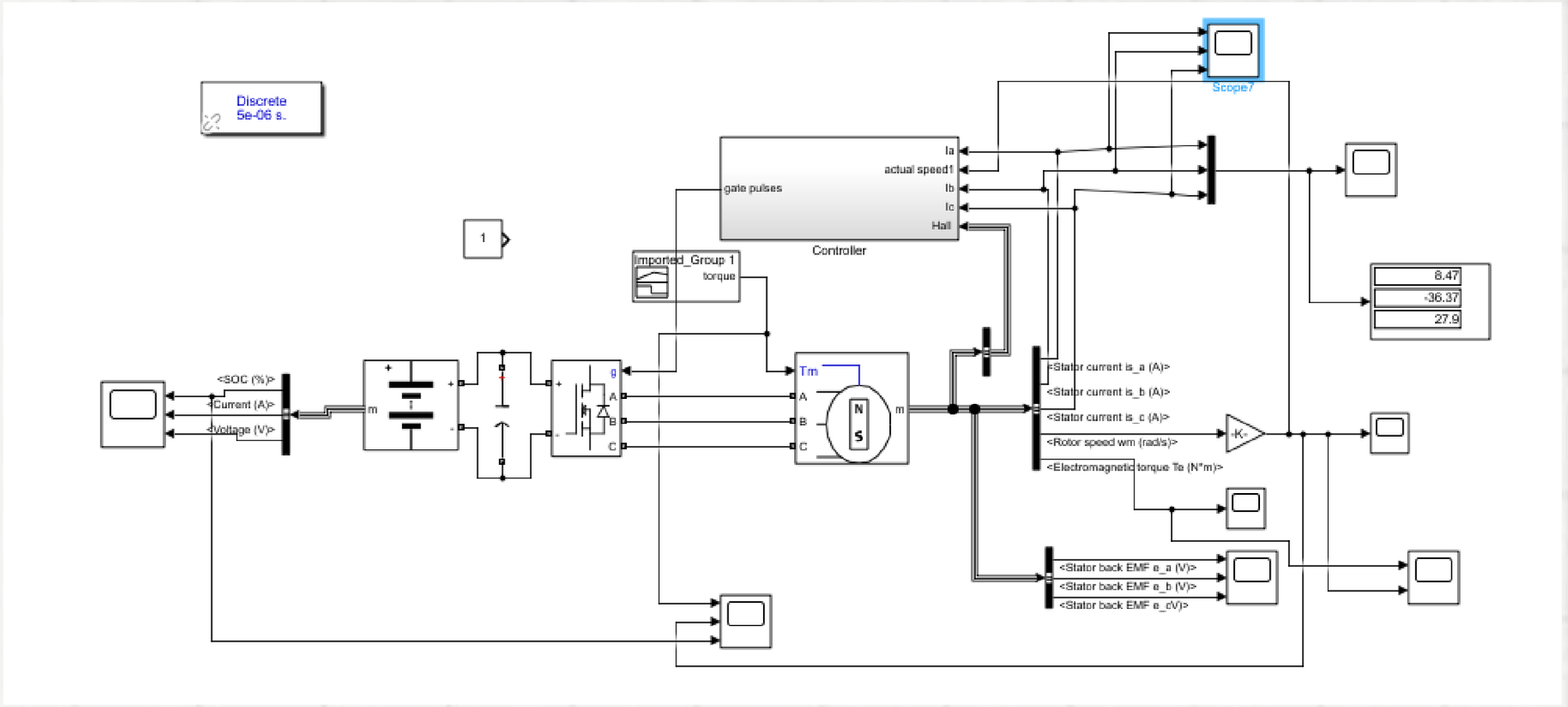


Why BLDC ?

- BLDC motors are high-speed motors and they also offer considerable torque.
- These motors are thus used in drones
- Back EMF is directly proportional to speed
- So, to get considerable back EMF, we use BLDC motor



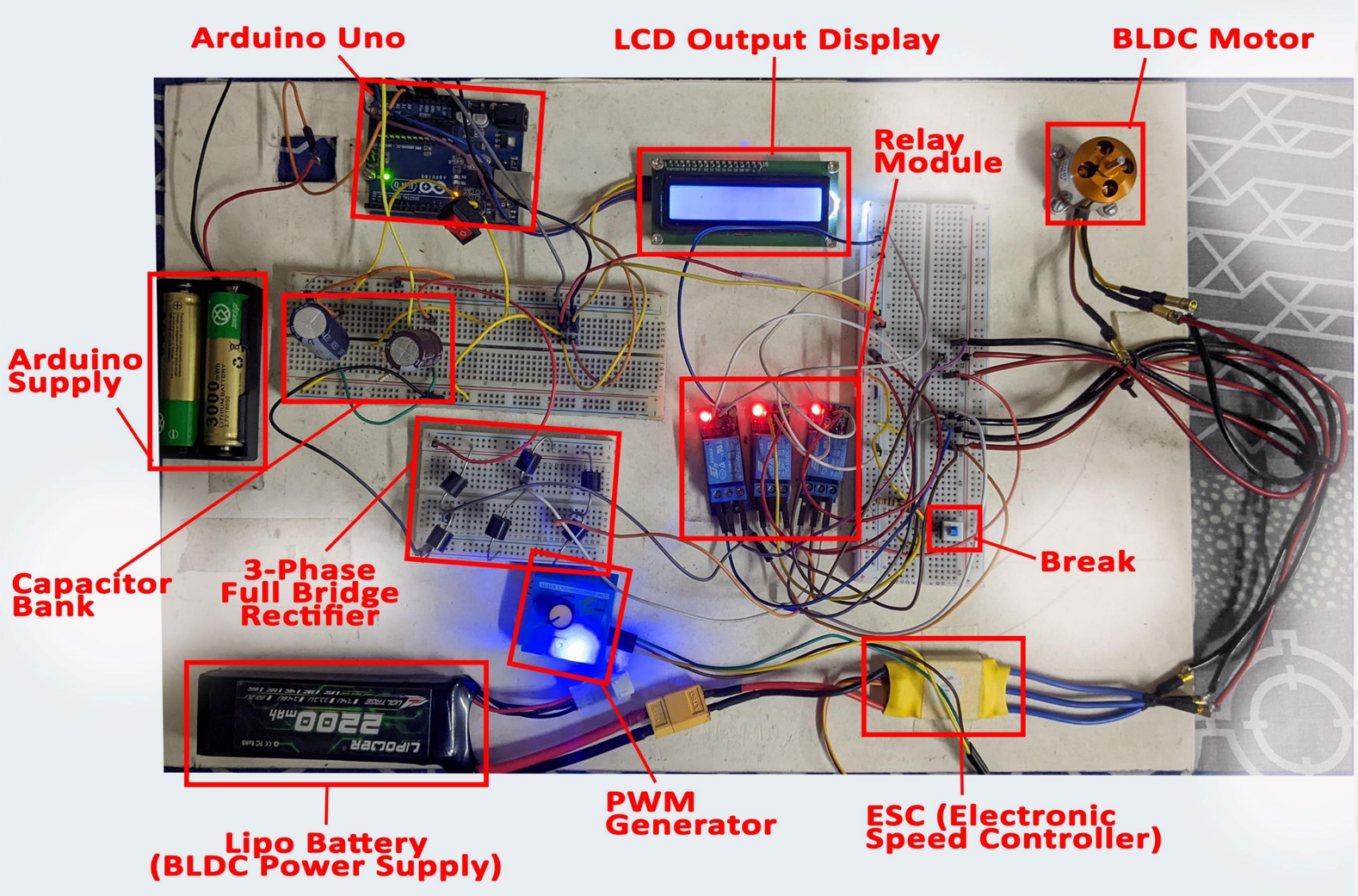
Simulink Model





Hardware Setup

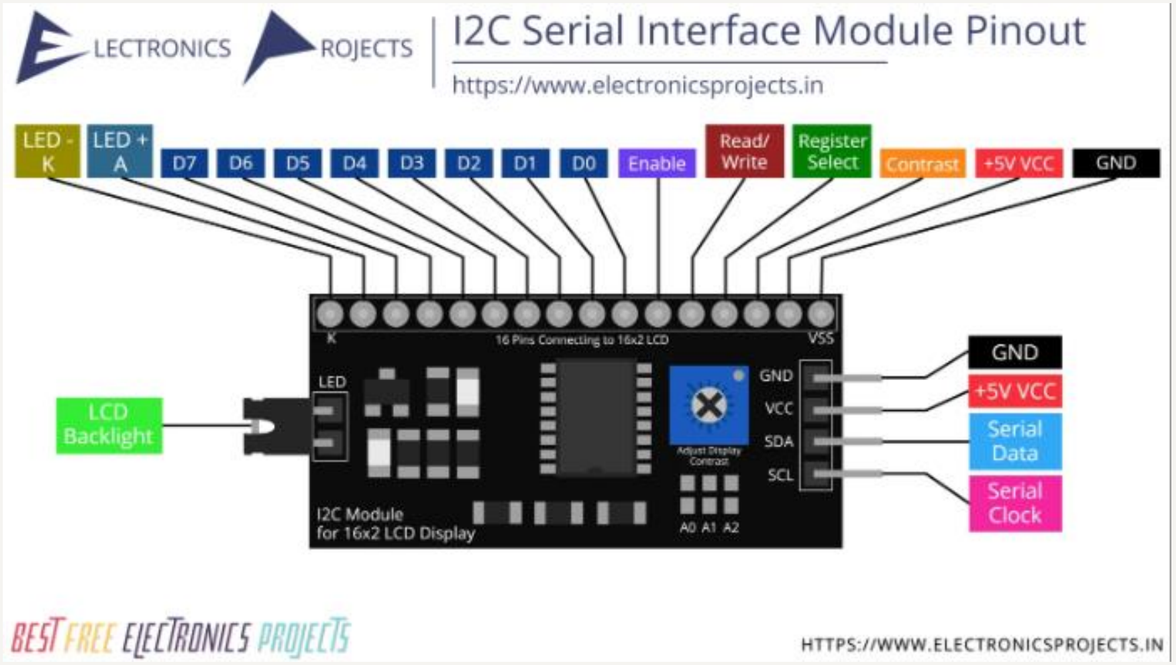
I2C Pinout

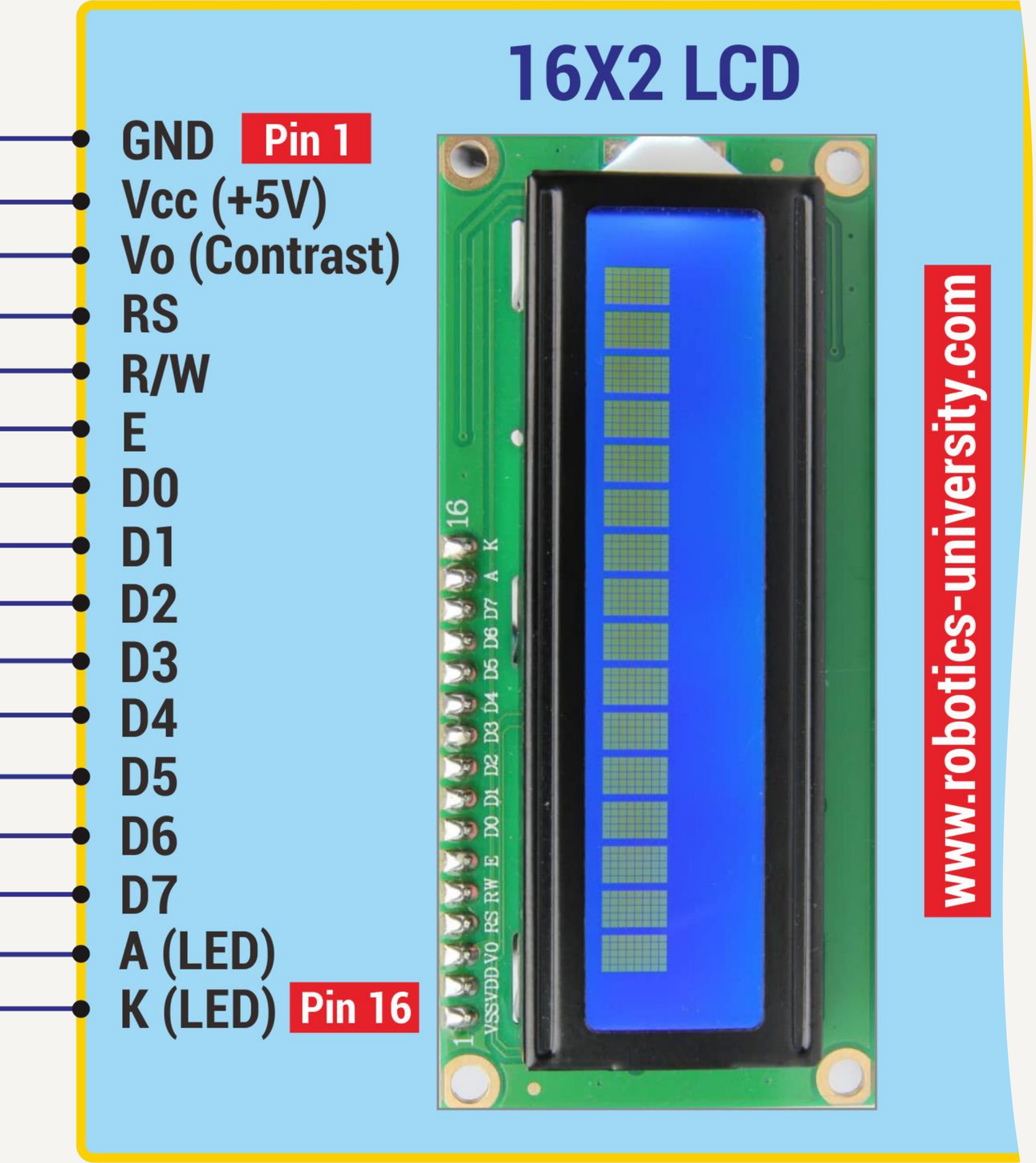




I2C

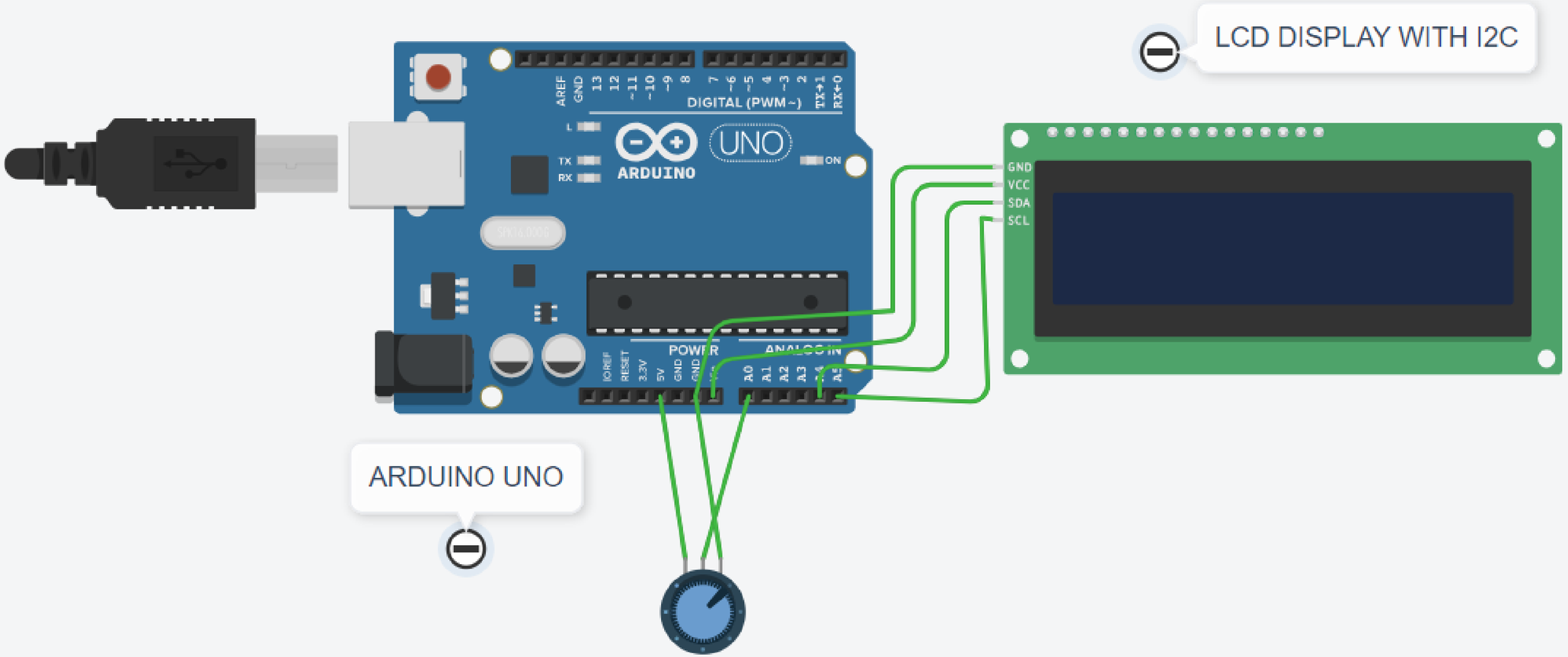
- The above 16 pins are dedicated for the LCD display
- SDA - Serial Data
- SCL - Serial Clock





LCD display

- **Contrast** pin
- **RS**(Register Select) pin
- **RW** pin
- **E** is enable pin
- **D0-D7** pins



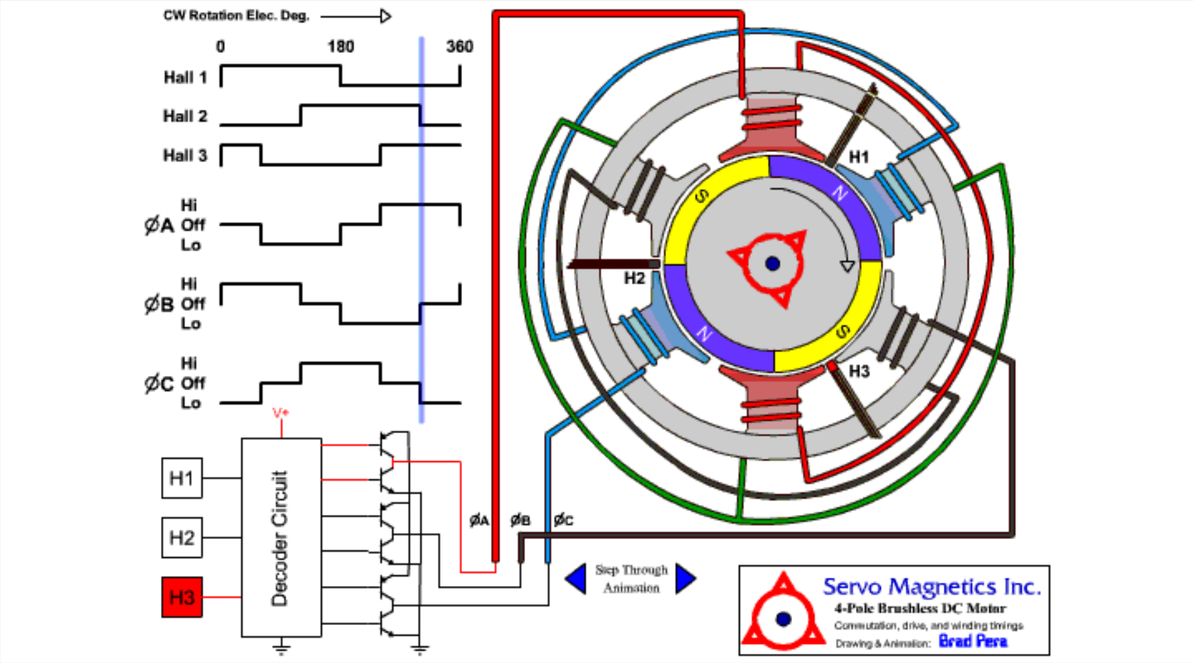
Circuit Diagram for LCD display

Why voltage divider is used at the input of the Arduino

```
void loop() {
  int analogValue = analogRead(analogPin); // Read analog input
  float voltage = (analogValue * referenceVoltage) / resolution*2; // Convert to voltage
```

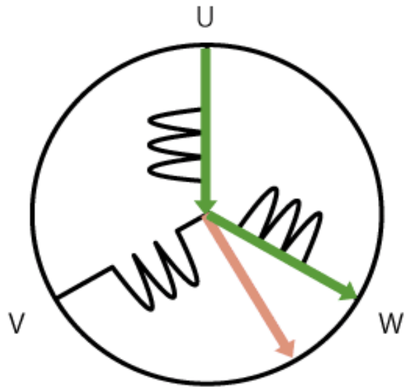
WE CAN'T GIVE MORE THAN 5 VOLT AS INPUT IN A0 PIN. BUT THE CAPACITOR GIVES ABOUT 6-7 VOLTS. TO DECREASE THAT VOLT WE USED VOLTAGE DIVIDER

How BLDC Motors Work?



- The stator has permanent magnets axially magnetized.
- Total Stator Poles= 14
- Total Rotor coils=12
- 12 coils divided into 3 sets, 4 coils per set
- Total Rotor phase=3
- Motor driver energizes the rotor coils sequentially to create net rotation .
- Outrunner motor- Stator is the outer casing and rotor is built within

How BLDC Motors Work?



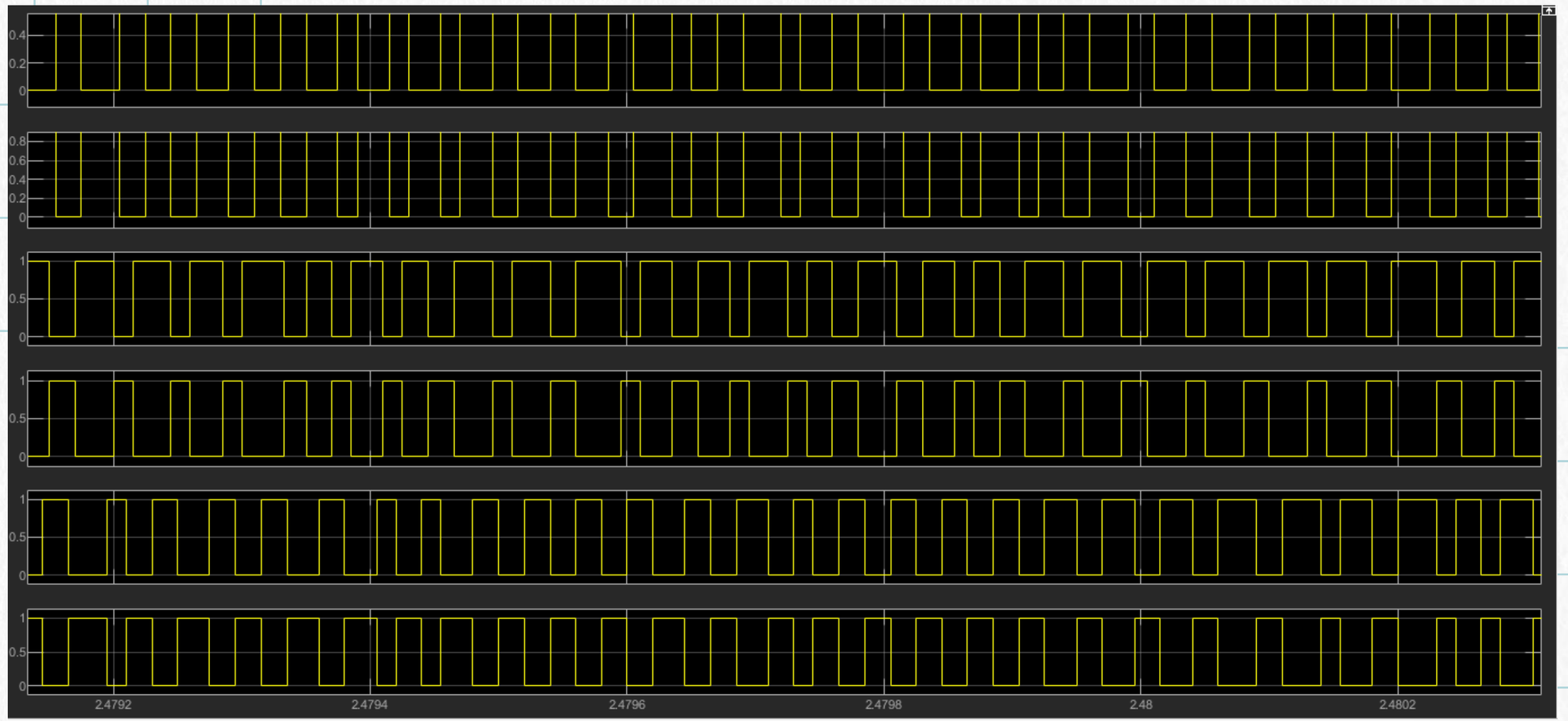
Energizing Mode	Energized Phase	Resultant Flux
1	U → W	
2	U → V	
3	W → V	
4	W → U	
5	V → U	
6	V → W	

r.

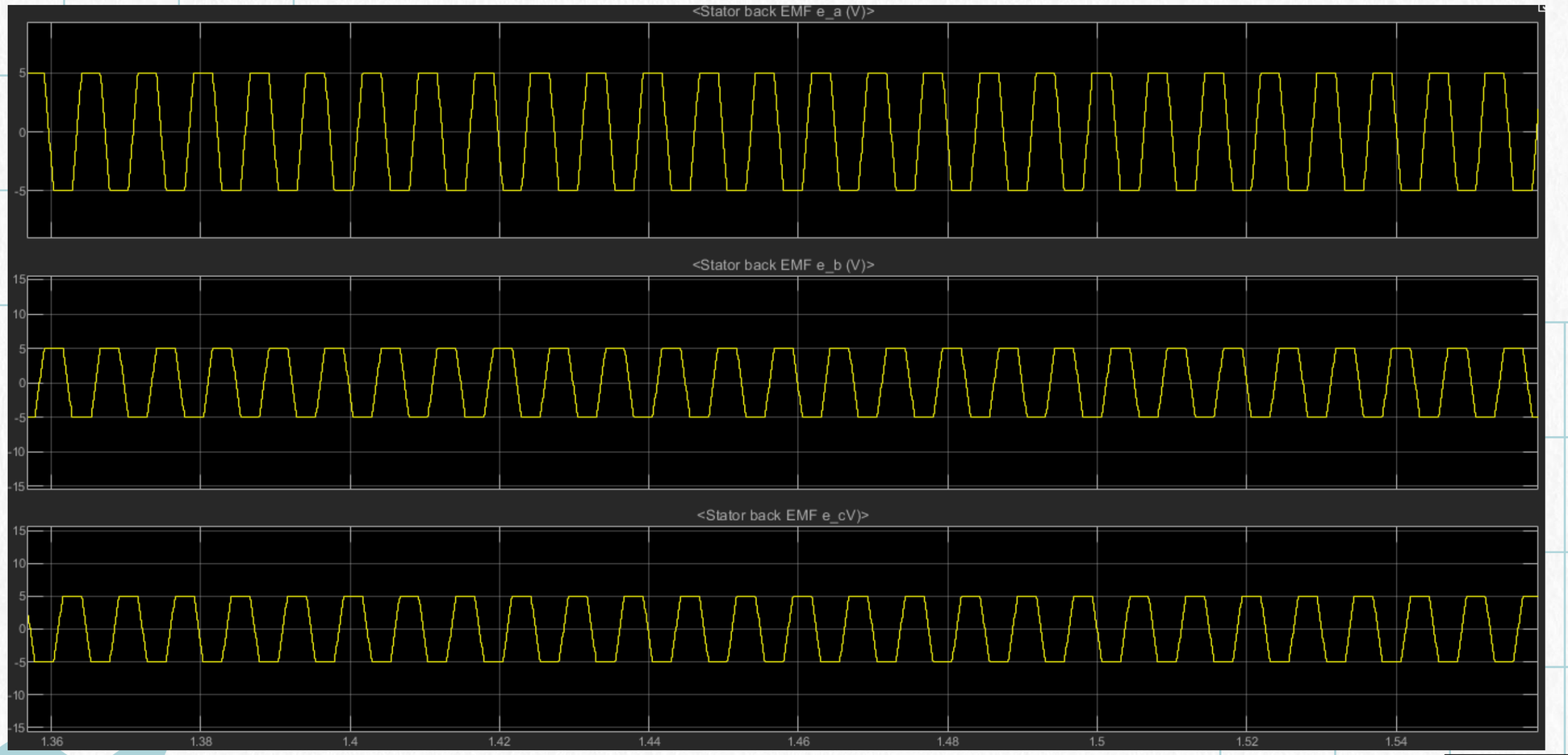
Rotor position (θ_e) (deg)	Hall sensor state			Switch closed		Phase Current		
	H1	H2	H3			A	B	C
0-60	1	0	0	Q1	Q4	+	-	off
60-120	1	1	0	Q1	Q6	+	off	-
120-180	0	1	0	Q3	Q6	off	+	-
180-240	0	1	1	Q3	Q2	-	+	off
240-300	0	0	1	Q5	Q2	-	off	+
300-360	1	0	1	Q5	Q4	off	-	+

Table 1: Six step switching sequence for Commutation

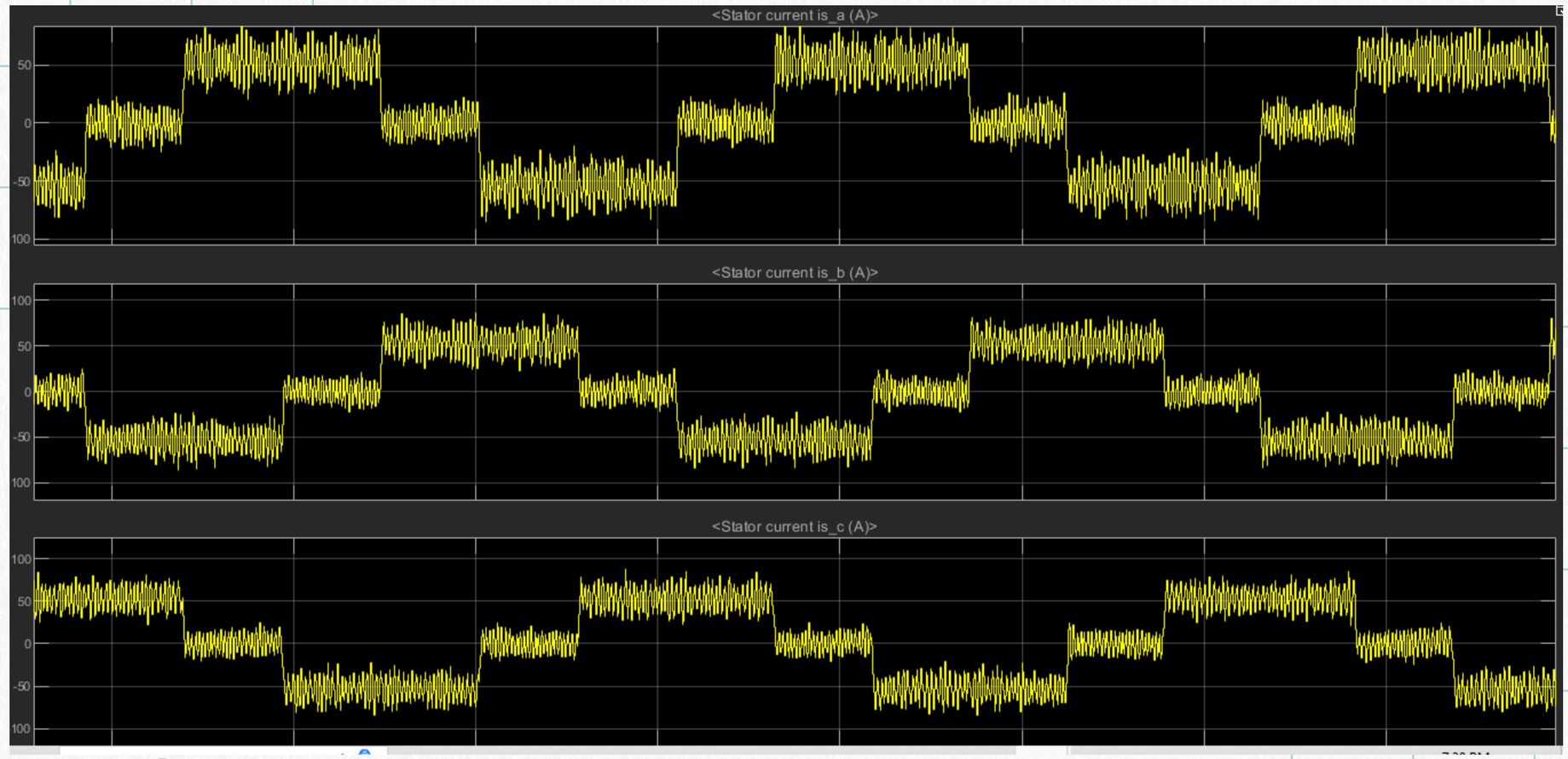
SIMULATION RESULTS: GATE PULSES



SIMULATION RESULTS: STATOR BACK-EMF



SIMULATION RESULTS: STATOR CURRENTS

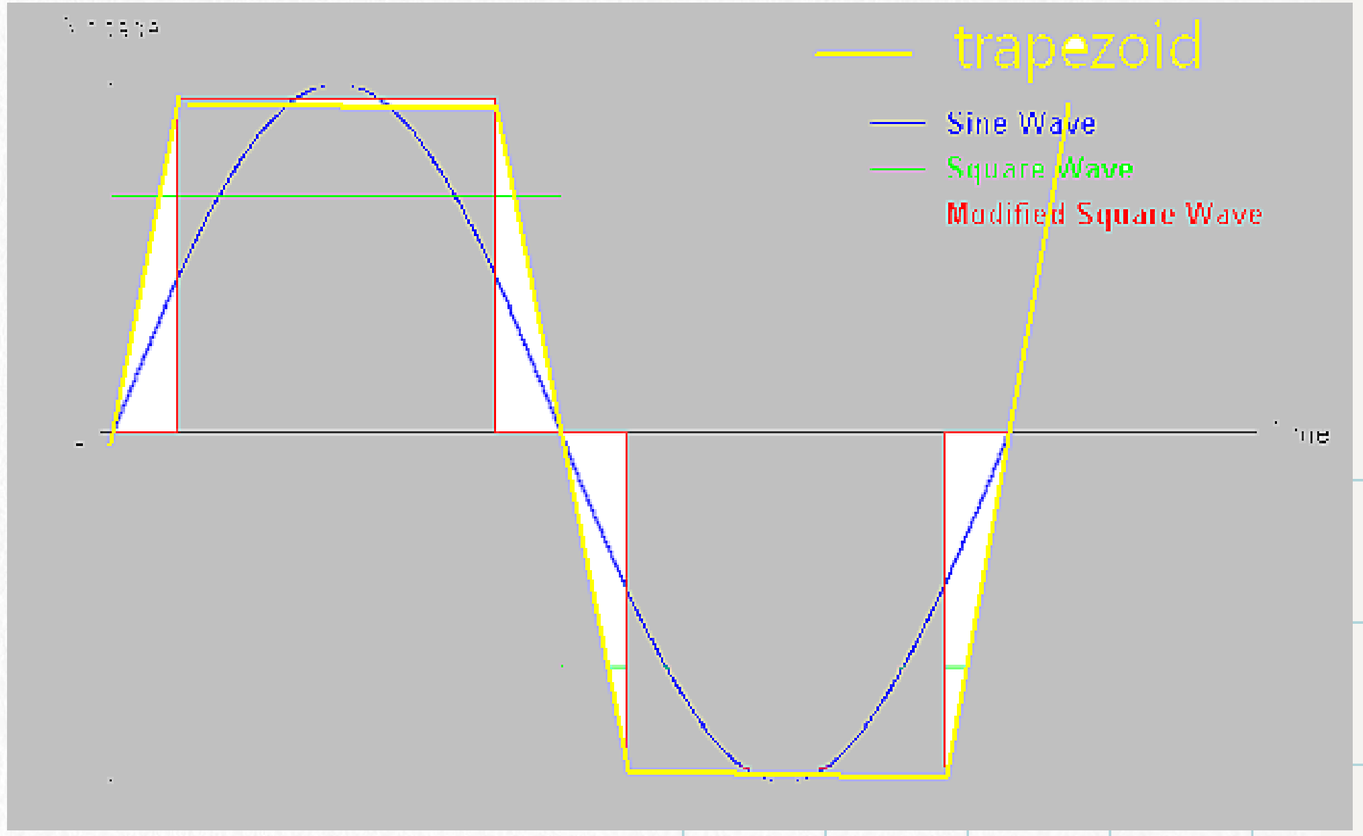


WHY TRAPEZOIDAL?

- The drive current should align with the back EMF waveform (trapezoid or sinusoid) for optimal performance.

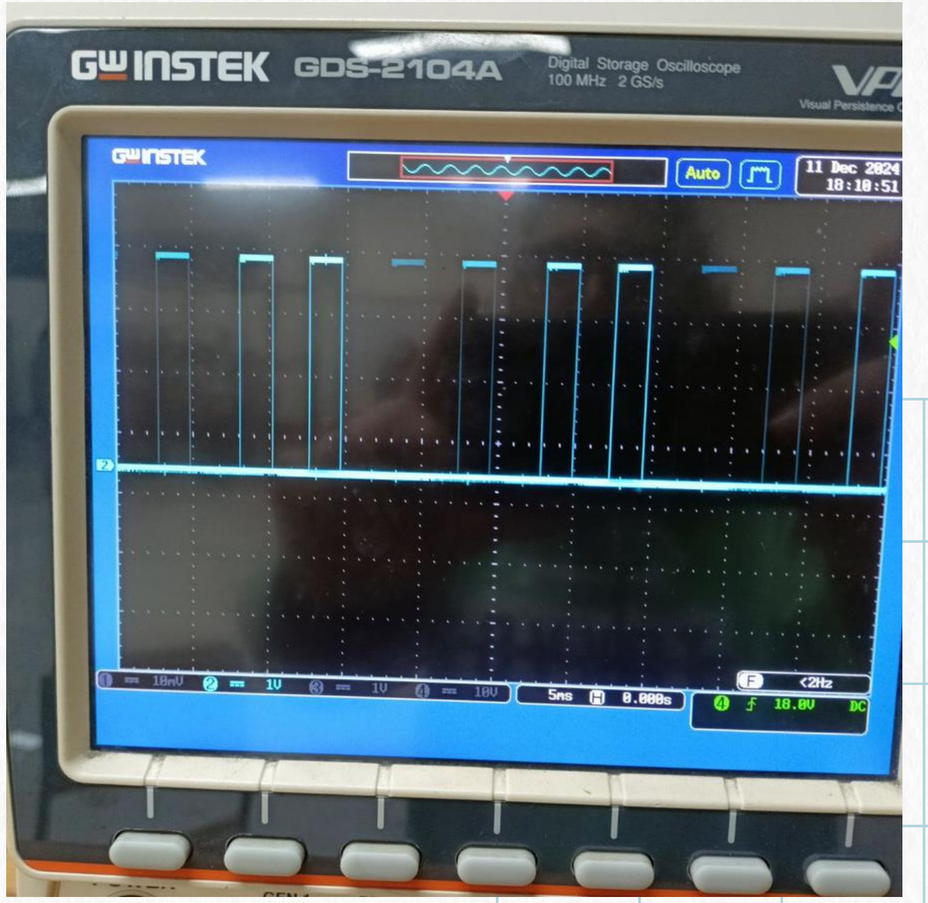
- As a rotor pole or moving magnet departs from the stator pole surface, the amplitude decreases linearly with the rotor's surface velocity.

- When the rotor pole surface overlaps with the stator magnetic field for a certain distance, the back electromotive force (BEMF) remains constant along that arc.



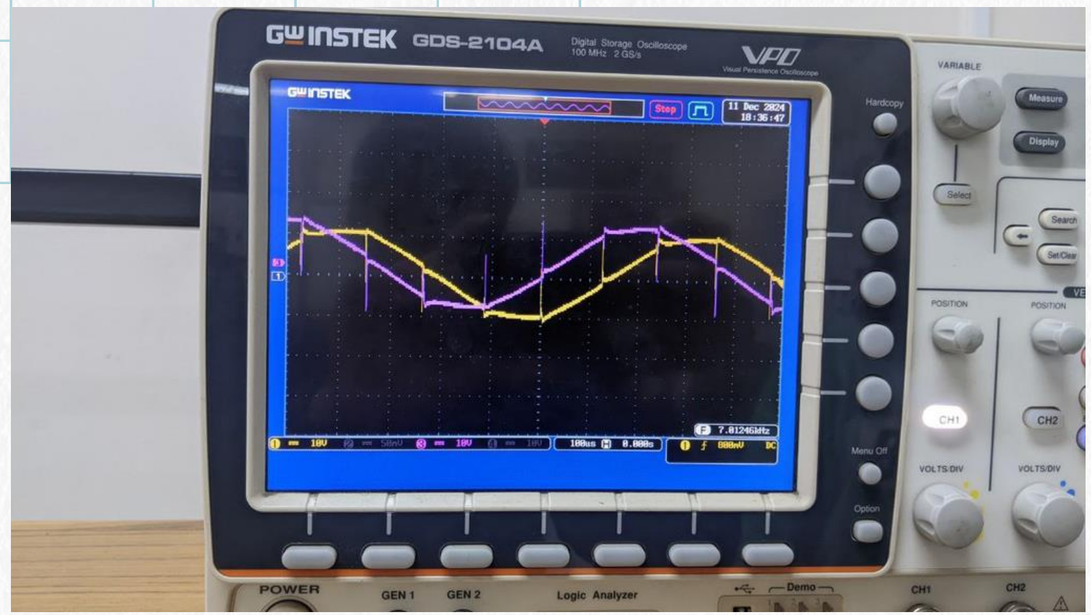
RESULT & FINDINGS

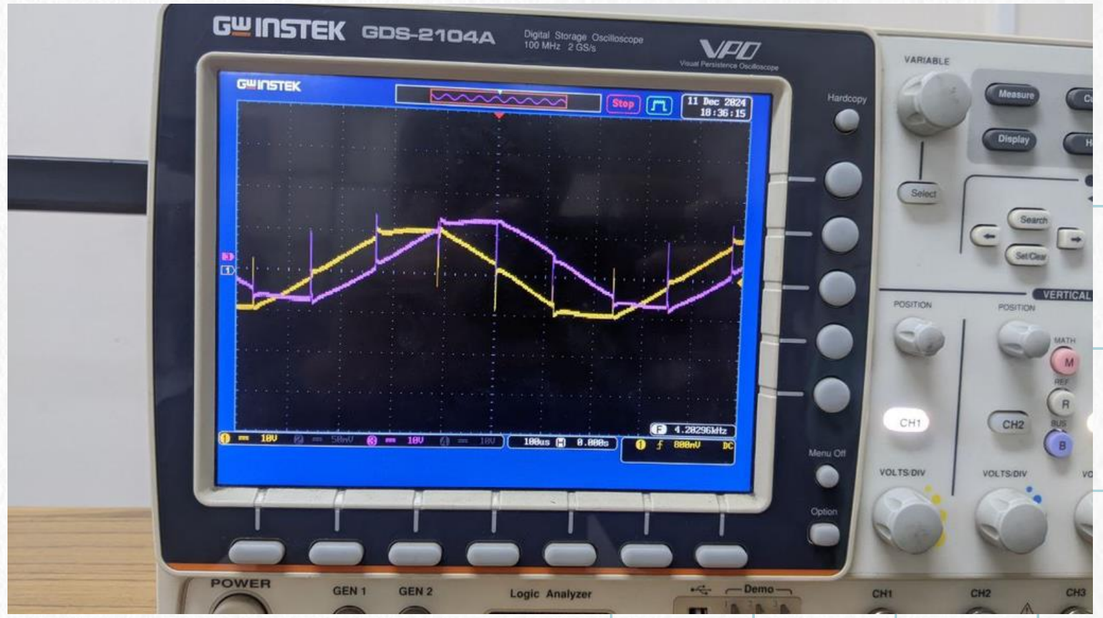
PWM Signal



RESULT & FINDINGS

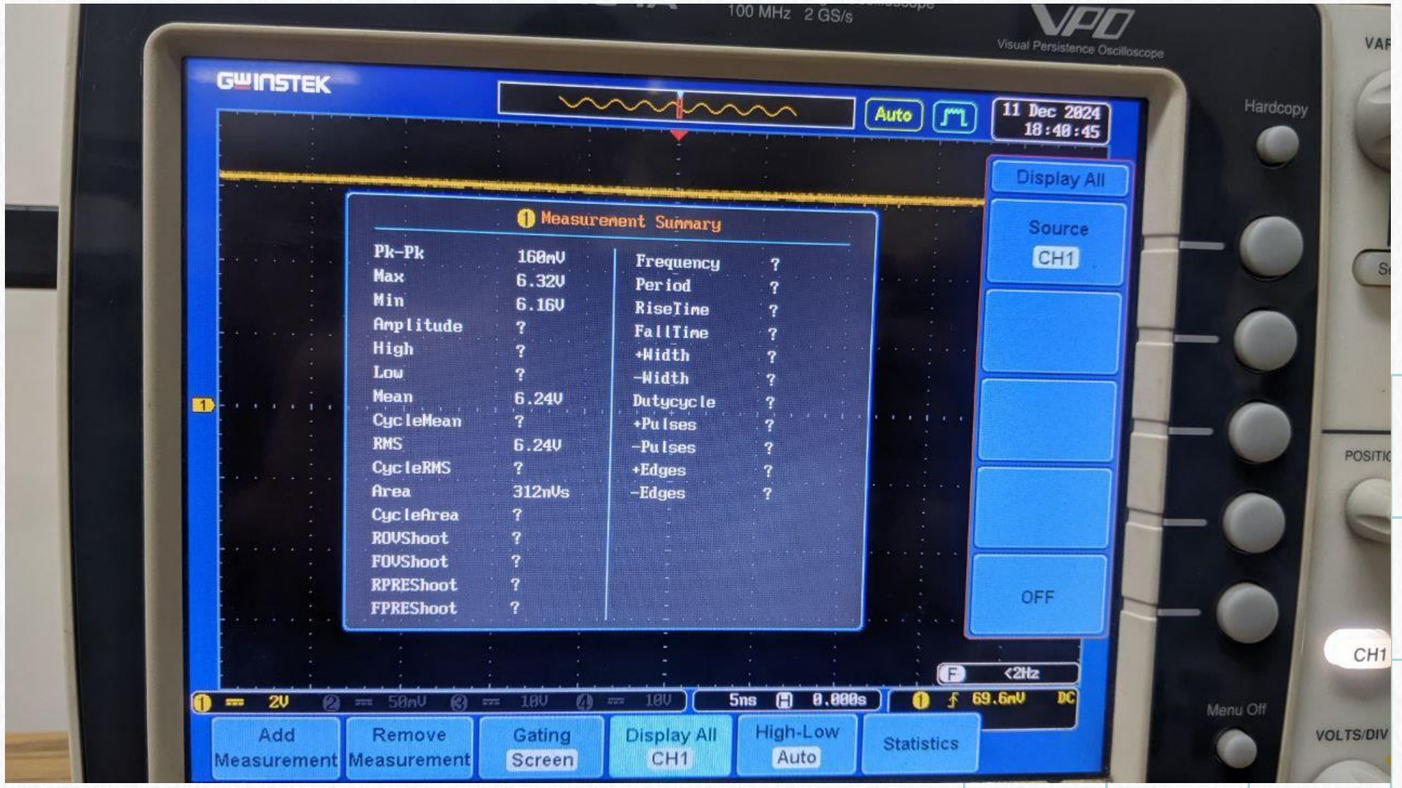
Motor Output Voltages(Line-Line)





RESULT & FINDINGS

Rectified Output :



FUTURE WORK

- Battery Integration
- Adaptive Control
- Energy Monitoring
- Scalability
- Bidirectional Operation
- Hardware Upgradation
- Renewable Energy Integration
- Thermal Management
- Rigorous Testing

COST ANALYSIS



Product Name	Quantity	Price
BLDC Motor(2212/10T-14kV)	x1	400
ESC30A	x1	450
Power Diode (1N5408)-1000V, 3A	x4	9x4=36
LCD Display-16x2	x1	110
Jumper Wire Box	x1	120
12AWG Cable	5 feet	30x5=150
Relay Module(5V)	x3	150
Breadboard	x7	75x7=525
PVC Sheet	x1	110
Servo Tester	x1	150
Capacitor=3300uF	x1	40
Capacitor-1000uF	x1	40
Battery- Li Ion(3.7V)	x2	140
Arduino Uno	x1	600
Subtotal		3021

REFERENCES

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- Jiyaad Lai, Lesedi Masisi, Udochukwu B. Akuru, "Regenerative Braking Torque Ripple Minimization of a Brushless DC Machine Using Fuzzy-PI Controller", *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, pp.6538-6540, 2023.

- Sumit Pradhan, Vineet Shekher, Mithlesh Kumar, "Regenerative Braking Control Strategy for Electric Vehicles based on PSO Optimization of BLDC Motor Speed Control", *2023 International Conference on Sustainable Communication Networks and Application (ICSCNA)*, pp.678-684, 2023.

THANK YOU VERY MUCH!